Introducing MathHelm: A Place to discuss AI-generated papers

Terence Tao · 13 November 2026

https://terrytao.wordpress.com/2026/11/13/introducing-mathhelm-a-place-to-discuss-ai-generated-papers/

tao2typ · converted to Typst 2026-11-14 (UTC)

[This is a guest post by the MathHelm scientific committee, whose membership is listed at the bottom of this post. -T.]

Mathematics depends on our regular conversations with colleagues and with the wider public. They help us share ideas, deepen our collective understanding, see where the field stands, and decide what to explore next. They also keep us connected. As technology changes rapidly, talking through uncertainty and supporting one another matter for our well-being and our shared effort to understand the universe.

On October 6, 2026, OpenAI released 722 mathematical manuscripts generated by an internal AI model, claiming significant results across many research areas. This release will undoubtedly bring significant changes to the mathematical community, even though most of these papers were not verified by us humans and among them, many results did not have a companion lean-verification.

Like many others, I was interested in several of these papers. I also found the whole collection intimidating. Their presentations often differ from the scholarly conventions I knew, and I could not quickly assess most of the arguments on my own. It was natural that I wanted to discuss with colleagues, but often felt hesitant to speak before I felt confident in my judgment. Sometimes I wasn't even sure whom to discuss with. A few colleagues felt the same way. Together, we created MathHelm: a place for everyone to share comments, ask questions, and think together on massive AI-generated papers, including these 722 papers and any other similar paper.

Each of us only has the expertise to assess a small part of such a broad collection. Useful discussions were already appearing online: explanations of an idea, questions about a proof, or missed connections to earlier or ongoing work and related concerns. People's comments are scattered and easy to miss. Bringing them together saves time for everyone and gives the community a clearer picture.

MathHelm is open to researchers worldwide, with discussions primarily in English. We invite mathematical researchers to register under their real names and discuss AI-generated papers and related mathematical questions. Comments can address correctness, readability, originality, main results, or key ideas. We hope everyone could be as objective and specific as possible, and include references where appropriate. You can also upload your own PDF notes.

The forum was founded by the scientific committee listed below and operates on a nonprofit basis to serve the mathematical community. It focuses on mathematical content and takes no predetermined position for or against AI. Our policy prohibits any organization from using forum content to train AI models.

Our guiding principle is simple:

Mathematicians have no obligation to check AI-generated papers. But if you happen to read part of one, your observations may be valuable to the whole community.

Even small comments such as a question about the proof method, an explanation of a single step, or a useful reference can help. You do not need a verdict on an entire paper to join the conversation.

Whatever tools we use, mathematics should remain a shared human pursuit: people should be able to understand its ideas, question its claims, and help shape its future. We hope MathHelm can offer a little steadiness in this age of rapid change, and a place to keep learning from one another.

MathHelm Scientific Committee

Xinyi Yuan (Peking University), Chair

Yi Ni (California Institute of Technology)

Xiaoming Sun (Institute of Computing Technology, Chinese Academy of Sciences)

Liang Xiao (Peking University)

Yuan Yuan (Westlake University)

Ruixiang Zhang (University of California, Berkeley)

Fan Zheng (ICMAT–CSIC, Spain)

Comments (2)

Anonymous · 13 November 2026

Readers may also be interested in the following related efforts:
* open-ai-revised: https://terrytao.wordpress.com/2026/10/24/oai-math-revised-a-community-hub-for-revising-openais-math-write-ups/
* Proofs and People: https://proofsandpeople.org/
* Zulip forum for Proofs and Prompts (not people): https://proofsandprompts.zulipchat.com/login/

By the way, I wonder if it would not make more sense for the AMS or similar organizations to host a service? This would help to avoid the fragmentation into many different forums, and give the ultimate control to the community.

I am a bit uncomfortable with the fact that all existing venues are run by unaccountable smallish groups. What can the community do if the organizers make controversial decisions? (e.g., unfair moderation when they clearly have a conflict of interest.) It would be nice if there was some higher power that could intervene.

I do not doubt that the organizers of these initiatives have the best intentions. Moreover, it makes sense that the day-to-day should be run by a small group that can decisive action. However, I would like some ultimate accountability to the community.

domotorp · 13 November 2026

“Our policy prohibits any organization from using forum content to train AI models.” There is absolutely no way to enforce that, and I’m not even sure if it is legal at all to make such a policy.